

Implementing a Software Defined Network (SDN) Based Firewall

Get the feel of software defined networking (SDN) — an approach to computer networking that allows admins to manage services by abstracting higher-level functionality. This article explains how to implement an SDN based firewall using SDN controllers in Python.

A firewall is a system that secures incoming network packets, which come from various sources, as well as outgoing network packets. It can monitor and control the flow of data which comes into the network from different sources, and works on the basis of predefined rules.

Firewalls typically maintain a barricade between a confidential, protected internal network and another outside network, such as the Internet, which is assumed not to be secure or trusted. They can be categorised as either hardware or software firewalls. Network firewalls are software programs running on different hardware appliances in the network.

Software based firewalls provide a layer of software on a host, which controls network traffic in and out of that particular machine. Firewall appliances may also provide other functionality to the internal network they protect, such as acting as DHCP or VPN servers for that network.

The system analyses data packets for parameters like layer2 or layer3 switch packet formats. It can also perform deep packet scrutiny for higher layer parameters (like application type and services, etc) to filter network traffic.

Firewalls are an essential component of any secure network communication for bi-directional packet flow.

Software defined networking

Software defined networking (SDN) is the new network technology. It emphasises the separation of the network and the control plane. Responsibility is divided between both the planes. The forward plane is only responsible for packet forwarding in the network. The control plane is responsible for policy creation and its implementation, based on predefined rules. It acts as the entry point of the system and can replace the conventional router. Any packet coming into the network is examined by the control plane and a decision is taken on whether to drop the packet or forward it to the next host. It can also update the IP table entry.

Hence, an SDN is the physical separation of the control plane from the data plane. So, instead of each networking device independently forwarding packets to the next hop, the controls are centralised on SDN controllers.

SDN is an engaging platform for network virtualisation, since each occupant's control logic can run on a controller rather than on physical switches. It is an approach to computer networking that allows network administrators to manage network services through the abstraction of higher level functionality.

So, SDN networks provide flexibility, programmability and simplicity to network operations. Traffic can be directed, adjusted or personalised without requiring physical wiring changes.

An SDN promises consolidated control and traffic management, which deals with automated network security that is more adaptive and mountable.

The characteristics of SDN are:

- Directly programmable
- Agile

